

GabayPoz Recommender System — Version 1

Demonstration of the Six-Phase Pipeline

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Abstract

This document is an executable demonstration of the GabayPoz recommender, version 1. It is generated by running `demo.py` against the bundled sample fixtures under `data/sample/` and emitting a LaTeX fragment (`results.tex`) that this wrapper includes verbatim. Each subsection describes one synthetic student profile, the inputs the recommender received, the top-three recommendations it produced, and a deep dive on the rank-1 pick: matched dimensions, applied constraints, scholarship context, the PIDS Graduate Tracer Study employment outlook, and the hedged explanation text rendered to the student. Rebuilding the PDF reproduces this content bit-for-bit; see `Makefile`.

1. About this demonstration

GabayPoz is a constraint-aware, explainable college program recommender for Grade 11/12 students in Pozorrubio, Pangasinan. Two rounds of EDA established that the binding constraints on tertiary access are not academic preparation but *geography* (mean barangay-to-university road distance: 76.9 km / 79.2 min) and *household economics* (a $12.7\times$ education-spending gradient across FIES income deciles). The recommender honours those findings by applying hard mobility and economic filters *before* any program-fit scoring takes place, and by emitting a structured 20-field explanation alongside every recommendation.

The pipeline runs in six phases:

1. **Student profile.** Q1–Q9 build a six-dimensional affinity vector $S \in [0, 5]^6$ over (STEM, HEALTH, ARTS, BUSINESS, EDUCATION, AGRICULTURE), with weights $s_d = 0.50 \text{ internal}_d + 0.40 \text{ aptitude}_d + 0.10 \text{ market}_d$. The market component is held flat at 1.0 in version 1 (TDS §4.5).
2. **Hard filters.** Q11 (commute-time ceiling per student-barangay $\rightarrow$ university) and Q10 (combined tuition-plus-transport burden against an FIES-anchored affordability tier) are applied to the program-school universe. A three-step relaxation cascade (Q11-only, Q10-only, both) precedes any `NO_CANDIDATES` error.
3. **Fit scoring.** A blended base = $0.70 \cos(S, P) + 0.30 (S \cdot P)/150$ favours directional alignment in the cold-start regime.
4. **Adjustments.** Q12 duration / board-exam soft penalty ($\delta = 0.85$ when program tolerance level exceeds the student's), then CHED / RDC priority-program boosters ($\beta = 1.10 \mid 1.10 \mid 1.18$ capped — never 1.21).
5. **Diversity-enforced ranking.** Greedy top-3 with a 0.15 score-gap threshold, so when ranks 1 and 2 fall in the same dominant dimension, the rank-3 slot opens to a different field unless the score gap is large.
6. **Explanation.** A 20-field structured JSON with matched dimensions, applied constraints, penalties and boosters, scholarship context (including the `mobility_cuts_scholarship` interaction flag), the PIDS GTS employment outlook, the `agriculture_compound_flag`, the

`low_confidence_flag`, and a hedged human-readable text. Field-derivation rules and the schema in full are in `docs/tds.md` §6.

The four profiles below were chosen to exercise each of the distinctive behaviours: the GTS employment outlook on a HEALTH recommendation (D1), the mobility-scholarship interaction (D2), the agriculture compound-disadvantage flag (D3), and the cumulative constraint-relaxation cascade (D4).

2. Demonstrative profiles

D1: Health-oriented, financially comfortable, ready for board exam

A Pozorrubio centre-barangay student with strong HEALTH affinity, willing to dorm and financially comfortable. Q12=A signals readiness for the board-exam pipeline.

Inputs.

- Barangay id: 1
- Q7 strand: STEM; Q10: C; Q11: C; Q12: A
- Q1–Q6 (interest, single-letter dim code): H H H H H H; Q8–Q9 (aptitude): H H

Top-3 recommendations.	Rank	Program	University	Final	Base
	1	BS Nursing	SLU Baguio	0.752	0.683
	2	BS Nursing	UPang Dagupan	0.752	0.683
	3	BS Computer Science	PSU Lingayen	0.180	0.164

Rank-1 explanation.

- Student vector: [0.10, 4.60, 0.10, 0.10, 0.10, 0.10]
- Program vector: [2.00, 5.00, 1.00, 0.50, 1.00, 0.50]
- Matched dimensions: HEALTH
- Constraints: commute 180 min (`q11_result=pass`); burden PHP 180000 at student tier 5 (`q10_result=pass`).
- Scholarships: `nearby=1`, `outside=2`, `mobility_cuts_scholarship=False`.
- GTS employment outlook for HEALTH: $P(\text{employed}) = 0.728$, in-field match = 0.49, median months to first job = 14.

Generated explanation text (rank 1).

Your responses suggest alignment with HEALTH-oriented programs. BS Nursing at SLU Baguio may be a good fit and passed your financial and travel preferences. It is a CHED priority program. 1 scholarship(s) may be available at this school. Nationally, 72.8% of Health Sciences graduates are employed, with a median of 14 months to first job.

D2: Arts-oriented, mobility-constrained — surfaces scholarship tradeoff

Strong ARTS affinity but Q11=A (nearby only). Demonstrates how the mobility filter interacts with scholarship geography: an Arts scholarship that exists only at outside schools is invisible to this student, and the explanation surfaces the tradeoff.

Inputs.

- Barangay id: 1
- Q7 strand: HUMSS; Q10: B; Q11: A; Q12: B
- Q1–Q6 (interest, single-letter dim code): A A A A A A; Q8–Q9 (aptitude): A A

	Rank	Program	University	Final	Base
Top-3 recommendations.	1	AB Communication	PSU Lingayen	0.690	0.690
	2	BS Business Administration	PSU Lingayen	0.320	0.291
	3	BS Computer Science	PSU Lingayen	0.180	0.164

Rank-1 explanation.

- Student vector: [0.10, 0.10, 4.60, 0.10, 0.10, 0.10]
- Program vector: [0.50, 0.50, 5.00, 2.00, 1.00, 0.50]
- Matched dimensions: ARTS
- Constraints: commute 40 min (q11_result=pass); burden PHP 55000 at student tier 3 (q10_result=pass).
- Scholarships: nearby=0, outside=1, mobility_cuts_scholarship=True.
- GTS employment outlook for ARTS: P(employed) = 0.717, in-field match = 0.72, median months to first job = 5.

Generated explanation text (rank 1).

Your responses suggest alignment with ARTS-oriented programs. AB Communication at PSU Lingayen may be a good fit and passed your financial and travel preferences. Nationally, 71.7% of Arts & Humanities graduates are employed, with a median of 5 months to first job.

D3: Agriculture-oriented — triggers compound-disadvantage flag

Strong AGRICULTURE affinity. EDA v2 documented that this field carries the sparsest scholarship coverage and the largest outside-only fraction. The recommender raises `agriculture_compound_flag` and surfaces the CHED+RDC priority offsets in plain language.

Inputs.

- Barangay id: 3
- Q7 strand: TVL; Q10: B; Q11: B; Q12: B
- Q1–Q6 (interest, single-letter dim code): A A A A A A; Q8–Q9 (aptitude): A A

	Rank	Program	University	Final	Base
Top-3 recommendations.	1	BS Agriculture	PSU Lingayen	0.859	0.728
	2	BS Civil Engineering	PSU Lingayen	0.165	0.165
	3	BS Medical Technology	PSU Lingayen	0.150	0.161

Rank-1 explanation.

- Student vector: [0.10, 0.10, 0.10, 0.10, 0.10, 4.60]
- Program vector: [1.00, 0.50, 0.50, 0.50, 0.50, 5.00]
- Matched dimensions: AGRICULTURE
- Constraints: commute 30 min (q11_result=pass); burden PHP 52000 at student tier 3 (q10_result=pass).
- Scholarships: nearby=0, outside=2, mobility_cuts_scholarship=False.
- GTS employment outlook for AGRICULTURE: P(employed) = 0.765, in-field match = 0.62, median months to first job = 5.
- **Agriculture compound-disadvantage flag set.**

Generated explanation text (rank 1).

Your responses suggest alignment with AGRICULTURE-oriented programs. BS Agriculture at PSU Lingayen may be a good fit and passed your financial and travel preferences. It is a CHED priority and RDC Region 1 priority program. Nationally, 76.5% of Agriculture & Vet Med graduates are employed, with a median of 5 months to first job. Note: Agriculture and Veterinary Medicine programs have fewer scholarships available nearby — 26 of 40 scholarships cover this field, and most are accessible only at outside schools. However, CHED and/or RDC Region 1 priority status may open additional scholarship windows. Please check DOST and DA scholarship periods directly.

D4: STEM-oriented, remote barangay, tight budget — fallback relaxation

Strong STEM affinity from a remote barangay (longer commutes) with Q10=A and Q11=A. The original constraint envelope yields no candidates; the three-step fallback cascade relaxes Q11, then (if still empty) Q10, and finally both simultaneously. The explanation flags both `*_constraint_relaxed` fields so the student is never silently moved to a different envelope.

Inputs.

- Barangay id: 2
- Q7 strand: STEM; Q10: A; Q11: A; Q12: B
- Q1–Q6 (interest, single-letter dim code): S S S S S S; Q8–Q9 (aptitude): S S

Top-3 recommendations.

Rank	Program	University	Final	Base
1	BS Computer Science	PSU Lingayen	0.777	0.706
2	BS Computer Science	UPang Dagupan	0.777	0.706
3	BS Medical Technology	PSU Lingayen	0.394	0.422

Rank-1 explanation.

- Student vector: [4.60, 0.10, 0.10, 0.10, 0.10, 0.10]
- Program vector: [5.00, 1.00, 1.00, 1.00, 1.00, 0.50]
- Matched dimensions: STEM
- Constraints: commute 75 min (q11_result=pass); burden PHP 72000 at student tier 3 (q10_result=pass).
- Constraint relaxation triggered: mobility, economic.
- Scholarships: nearby=2, outside=2, mobility_cuts_scholarship=False.
- GTS employment outlook for STEM: P(employed) = 0.768, in-field match = 0.55, median months to first job = 7.

Generated explanation text (rank 1).

Your responses suggest alignment with STEM-oriented programs. BS Computer Science at PSU Lingayen may be a good fit and passed your financial and travel preferences. It is a CHED priority program. 2 scholarship(s) may be available at this school. Nationally, 76.8% of STEM graduates are employed, with a median of 7 months to first job.

3. How to reproduce this PDF

```
cd demo
```

```
make                # runs demo.py then pdflatex twice
```

`make clean` removes the LaTeX intermediates and the generated `results.tex`; `make distclean` additionally removes `demo.pdf`. Re-running from scratch reproduces the PDF deterministically: the recommender is fully stateless, the seven sample CSVs are version-controlled, and tie-breaking is documented in TDS §5 Phase 5.

4. References

- **TDS:** `docs/tds.md` — the authoritative specification this implementation traces to.
- **EDA v1:** `docs/eda_v1.pdf` — Pangasinan education, employment, and access profile.
- **EDA v2:** `docs/eda_v2.pdf` — Pozorrrubio program access, scholarship analysis, and employment signals.
- Burke, R. (2002). Hybrid recommender systems. *User Modeling and User-Adapted Interaction*, 12(4).
- Liu, T.-Y. (2009). Learning to rank for information retrieval. *Foundations and Trends in IR*, 3(3).
- Philippine Institute for Development Studies (2019). *Tracer study of CHED-scholarship and non-scholarship graduates* (Discussion Paper No. 2019-26).